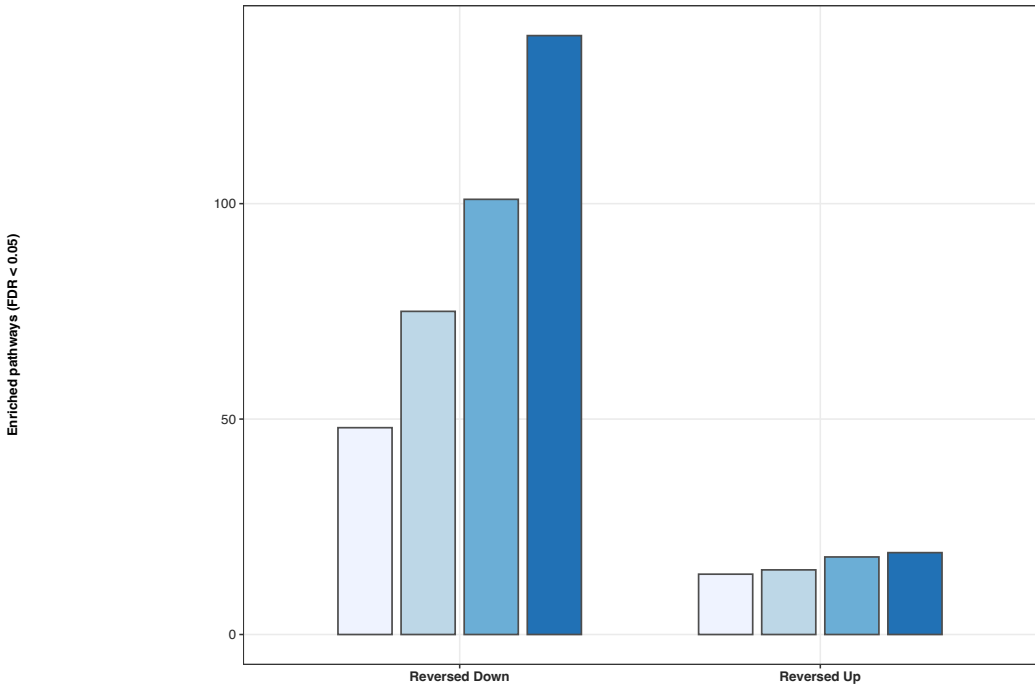
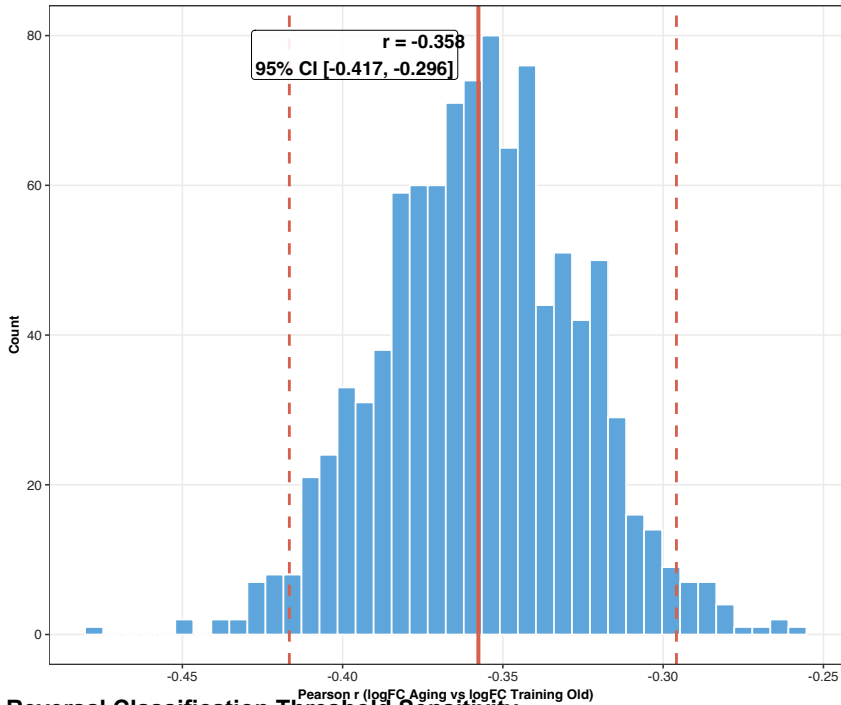


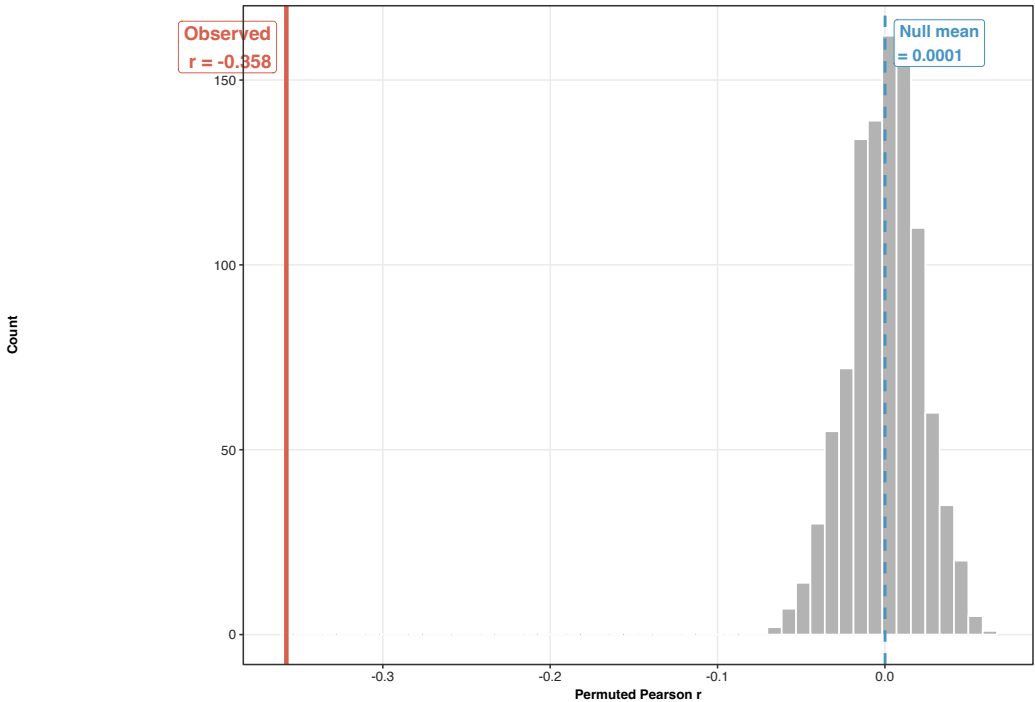
A **ORA Dedup Sensitivity (Reversal Quadrants)**
Hypergeometric ORA | 2103 universe | Reversed quadrants only



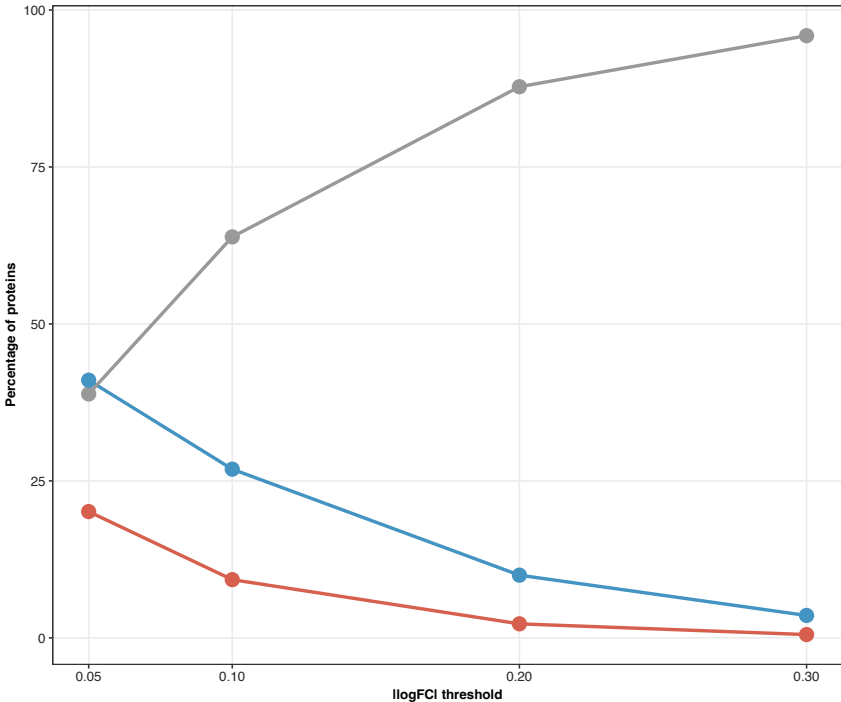
B **Reversal Pearson r Bootstrap**
boot::boot() R = 1000 | n = 2103 | r = -0.358 [-0.417, -0.296]



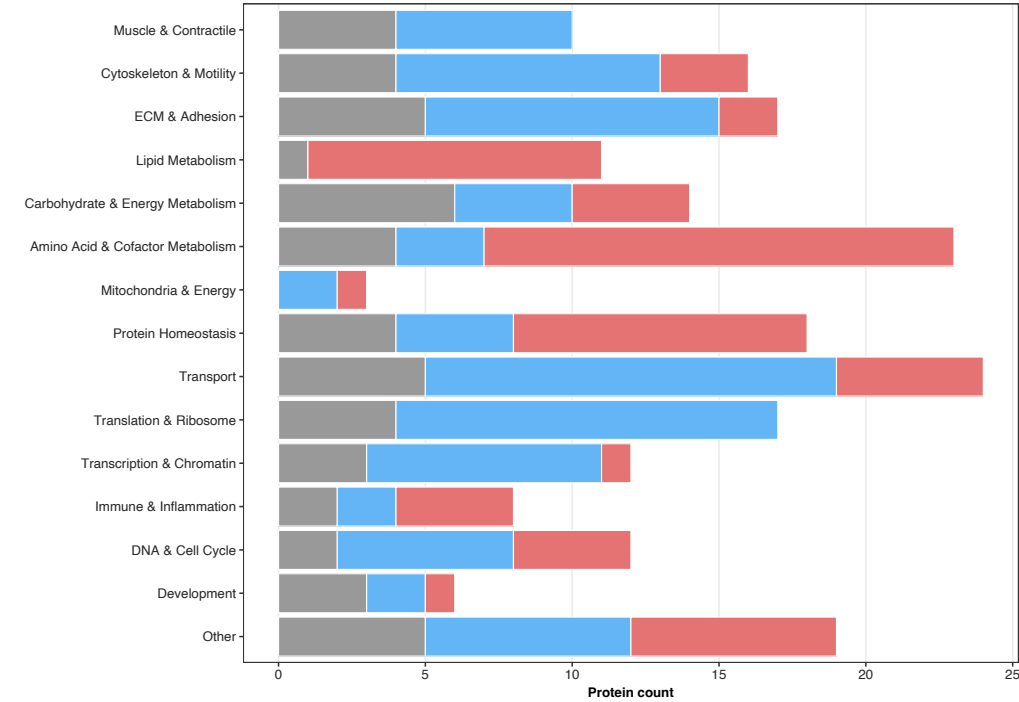
C **Circularity Diagnostic: Protein-Permuted Null**
Observed r = -0.358 | Null mean = 0.0001 | p_perm = 0.000



D **Reversal Classification Threshold Sensitivity**
logFC thresholds: 0.05, 0.1, 0.2, 0.3 | stable reversal %



E **GO Slim Category Distribution (Reversal)**
DEPs ($\Pi < 0.05$) | 210 proteins | 15 categories



F **fry Leading-Edge Proteins (Reversal Drivers)**
Top 25 by |t(Training Old)| | colored by aging direction

